



CSE356: Internet of Things Project Specifications

Objective:

Design and prototype an Internet of Things (IoT) system that addresses a specific problem or scenario. You can work in groups of 4-5 students on the project. The project is divided into three parts:

1. **Complete Design of an IoT System - (10 Marks)**
2. **Prototype Implementation Using Cisco Packet Tracer - (5 Marks)**
3. **Presentation and Discussion - (5 Marks)**

Part 1: Complete Design of the IoT System

Design Methodology: The students must follow the 10-step IoT design methodology outlined below. Each step should be well-documented and justified with respect to the chosen project scenario.

1. **Step 1: Purpose & Requirements**
 - Define the purpose of the IoT system.
 - List the functional and non-functional requirements.
2. **Step 2: Process Specification**
 - Detail the processes that the IoT system will perform.
 - Include diagrams or flowcharts if necessary.
3. **Step 3: Domain Model Specification**
 - Identify and describe the key entities and relationships in the domain of the IoT system.
 - Use UML diagrams to represent the domain model.
4. **Step 4: Information Model Specification**
 - Specify the data that will be used, generated, and managed by the IoT system.
 - Detail the structure and flow of information.
5. **Step 5: Service Specifications**
 - Define the services that the IoT system will offer.
 - Include descriptions of service interfaces and operations.
6. **Step 6: IoT Level Specification**
 - Specify the different IoT levels (e.g., sensing, network, application) and how they interact within the system.
7. **Step 7: Functional View Specification**
 - Develop a functional view that details the functional blocks and their interactions.
 - Include a description of how each functional block contributes to the overall system functionality.
8. **Step 8: Operational View Specification**
 - Outline the operational aspects of the IoT system, including performance, reliability, and security considerations.
9. **Step 9: Device & Component Integration**
 - Detail the devices, sensors, and actuators to be integrated into the IoT system.
 - Explain how they will be interconnected and communicate with each other.
10. **Step 10: Application Development**
 - Describe the application(s) that will interface with the IoT system.
 - Include user interface design and functionality, if applicable.

Deliverables for Part 1:

- A comprehensive report that includes all 10 steps of the design methodology.
- Diagrams, models, and any other relevant documentation to support the design decisions.



Part 2: Prototype Implementation Using Cisco Packet Tracer

In this part, students are required to implement a prototype of their IoT system using Cisco Packet Tracer.

Prototype Requirements:

- **Simulation of Key Components:** Students must simulate key components of their IoT system, such as sensors, actuators, networks, and data flows.
- **Configuration:** Properly configure the network elements (e.g., routers, switches) and IoT devices in Cisco Packet Tracer.
- **Testing:** Perform tests to ensure that the system operates as designed.
- **Documentation:** Document the simulation process, including the setup, configuration steps, and testing results.

Deliverables for Part 2:

- A fully functional Cisco Packet Tracer file containing the simulated IoT system.
- A brief report explaining the simulation, the challenges faced, and how they were overcome.

Part 3: Presentation and Discussion (5 Marks)

1. Time management
2. Presentation Clarity
3. Q and A.

Tools and links

- Search for Published Research
 - EKB
 - IEEE Xplore Digital Library
 - ACM Digital Library
- Figures and flowcharts
 - Gliffy
- Writing
 - Mendeley
 - Overleaf
- Spelling and grammar
 - Grammarly
- ScienceDirect
- SpringerLink
- Google Scholar
- Draw.io
- EndNote
- Ms Word

Submission Guidelines:

- **Due Date:** Wednesday, May 14, 2025
- **Discussion Date:** TBD
- **Format:** Submit the design report as a PDF and the Cisco Packet Tracer file via the university's submission portal
- **Assessment Criteria:** The project will be assessed based on completeness, adherence to the design methodology, functionality of the prototype, creativity, and quality of documentation.



PPT marking criteria

Course Code:					Course Name:					Assignment No.					Date:					
Student Name:											Student ID:									
	A (89-100)				B (76-88)				C (67-75)				D (60-66)				F (0-59)			
	100	96	92	89	88	84	80	76	75	72	69	67	66	64	62	60	59		20	0
Relevance & Organization of Ideas (50%)	- Materials cover the topic widely & deeply. - All main points fully developed. No repetition				- Materials cover the topic widely & deeply. - Clearly structured. All main points valid but not always fully developed. Minor repetition or deviation.				- Most of the materials cover the topic reasonably. - Some structure. Most but not all main points valid & developed. Some repetition.				- Some of the materials are relevant and slightly cover the topic. - Structure not clear. Few valid main points. Repetition or deviation.				- Materials do not cover the topic. - Unstructured. Few, if any, valid main points. Material mostly deviated from the task. Inaccurate or absent.			
Answering Questions (30%)	Answers are correct and to the point.				Most answers are correct.				Some answers are correct.				Few answers are correct.				Most answers are incorrect.			
Presentation Language (20%)	Excellent ability to express ideas with proper language and technical vocabulary.				Good ability to express ideas with proper language and technical vocabulary.				Normal ability to express ideas with proper language and technical vocabulary.				Low ability to express ideas with proper language and technical vocabulary.				Difficult to express ideas with proper language and technical vocabulary.			



Report marking criteria

Course Code:				Course Name:				Assignment No.				Date:								
Student Name:												Student ID:								
	A (89-100)				B (76-88)				C (67-75)				D (60-66)				F (0-59)			
	100	96	92	89	88	84	80	76	75	72	69	67	66	64	62	60	59	40	20	0
Literature survey (25%)	• Critical evaluation and synthesis of relevant issues and materials				• Critical evaluation of relevant issues and materials				• Accurate description of main relevant issues				• Limited evaluation and description of main issues				• Insufficient and largely irrelevant material			
Research Objectives (25%)	• Clearly defined research problem with well-structured research objectives				• Complete set of research objectives				• Limited research objectives				• Poorly defined objectives				• Research problem lacking clear objectives			
Research Methodology	• Clear and relevant research methodology with complete implementation				• Clear and relevant research methodology missing few components				• Clear research methodology missing several components				• Inappropriate research methodology				• Lack of clear research methodology			
Analysis of Results & Conclusions	• Excellent analysis of results and complete relevant conclusions				• Good analysis of results missing some minor conclusions				• Normal analysis of results missing some basic conclusions				• Incomplete analysis or results with some conclusions				• Missing proper analysis or results and no conclusions at all			

Assessment Method	LO1	LO2	LO3	LO4	LO5	LO6
Project	■			■		■

Best wishes
Dr. Islam Tharwat Abdel Halim